

Internet Safety, Responsible AI & Our First Algorithm

Digital citizenship • Trustworthy AI habits • Writing step-by-step instructions

TIER A • GR 1-3

TIER B • GR 3-5

TIER C • GR 6-9

HW DUE
Thu Jul 2

✓ TEACHER ANSWER SHEET — SESSION 2 OVERVIEW (Handout-1-2-Part-1)

Lesson Summary

♥ INTERNET SAFETY

- 1 Personal info** — never share name, address, or phone online.
- 2 Strangers** — online friends may not be who they claim.
- 3 Passwords** — keep secret, strong, never share.
- 4 Downloads** — always ask a trusted adult first.
- 5 Be kind** — posts last forever. Think before you post.

Accept any paraphrase that captures the core rule. Award 1 pt per rule.

⚙️ WHAT IS AN ALGORITHM?

✓ A precise, ordered set of step-by-step instructions a computer follows exactly — like a recipe with no missing steps.

Four properties: (1) Clear START & END (2) Right ORDER (3) Precise (no gaps) (4) Can use **IF/ELSE** and **WHILE** loops.

📱 RESPONSIBLE AI

- AI **can be wrong** — verify facts with a trusted source
- AI learns from data that may have **bias or mistakes**
- Never share **private info** with AI chatbots
- Use AI as a **thinking tool**, not a shortcut
- If AI seems wrong — **check a trusted source**

📖 KEY VOCABULARY

Algorithm	Precise step-by-step instructions ✓ "Like a recipe — ordered, precise, no gaps"
Debug	Find and fix an error ✓ "Remove the bug so the code works correctly"
IF / ELSE	Decision point: do A or B? ✓ "A fork in the road — IF yes → this, ELSE → that"
Loop (WHILE)	Repeat while condition true ✓ "Keep doing this step until something changes"
Private Info	Data only trusted people should see ✓ "Name, address, password — never share online"

🔗 In-Class Activity

🔥 WARM-UP • 10 MIN • ALL TIERS

→ **Puzzle 1:** Should you tell a robot your address?

✓ **NO. Address = private info. A robot/AI could store or misuse it. Rule: never share personal info with any program without a trusted adult's permission.**

→ **Puzzle 2:** What goes wrong with "Go to school"?

✓ **Too vague — a robot needs every detail: which direction? how far? what door? "Go to school" has gaps. Computers cannot guess; missing steps = algorithm failure.**

♥ PART 1 — SAFETY • 15 MIN

True/False Quiz Key:

- Passwords should use pet names. **F**
- Online friends are always real. **F**
- Be kind online as in person. **T**
- Ask adult before downloading. **T**
- Sharing address online is fine. **F**

3 Scenarios Key:

S1: App asks for real name + school

⚡ NOT SAFE — tell adult, close app. Personal info = private.

S2: Online friend asks where you live

⚡ NOT SAFE — stop chat, tell parent. Real friends don't need your address online.

S3: Mean comment on class chat

⚡ NOT SAFE — screenshot, tell adult, don't reply meanly. Online = real kindness.

Internet poster: accept any creative design with ≥2 rules shown clearly.

⚙️ PART 2 — ALGORITHM • 20 MIN • TIERED

🔗 TIER A — 4-Step Morning Routine

✓ e.g. Wake up → Brush teeth → Eat breakfast → Leave. Drawn in labeled boxes. Buddy swap: partner must follow it without guessing any step.

📅 Homework — Due Thu Jul 2

♥ SECTION 1 — SAFETY • ~5 MIN • ALL

✓ **5 Safety Rules: Personal info / Strangers / Passwords / Downloads / Be kind. Accept paraphrase. 1 pt each.**

T/F Quiz:

- Pet name = strong password. **F**
- Screenshot can be shared freely. **F**
- Click only trusted links. **T**
- Online kindness = real kindness. **T**
- Full name + address = private. **T**

✓ **3 Home Scenarios: S1 NOT SAFE (name+school). S2 NOT SAFE (address). S3 NOT SAFE (cyberbullying). See Activity col for full explanations.**

Stay Safe Online poster: ≥2 rules + title + drawn example = full credit.

⚙️ SECTION 2 — ALGORITHM • TIERED

🔗 TIER A • ~10 MIN

Draw & label 4-6 step home-task algorithm. Test with family!
✓ e.g. Making cereal: 1) Get bowl 2) Pour cereal 3) Add milk 4) Get spoon 5) Eat. Each step labeled. Family can follow without guessing.

💧 TIER B • ~12 MIN

6 steps + IF/ELSE + flowchart. Family debug — record & fix bug.
✓ e.g. Packing bag: 6 precise steps + **IF** homework done → pack it. **ELSE** do homework first. Bug: "I forgot to say which bag."

🔗 TIER C • ~15 MIN

8+ steps + 2 IF/ELSE + WHILE. Flowchart + pseudocode + reflection.
✓ Getting ready: **WHILE** alarm rings → snooze. IF >30 min → shower; ELSE wash face. IF bag packed → go; ELSE pack. Python pseudocode w/ correct indent. Reflection: 1 similarity between AI & algorithms.

📱 SECTION 3 — AI • ~5 MIN • ALL

AI always gives correct answers. **F**

AI Hallucination

AI invents a false but convincing answer

- ✓ "AI sounds confident but fact is wrong — always verify!"

TIER B — 6 Steps + IF/ELSE + Flowchart

- ✓ Steps are specific (which hand, how long). **IF** alarm didn't ring → check time. **ELSE** proceed.
- Flowchart: diamonds=decisions, rectangles=actions. Bug recorded.

TIER C — 8+ Steps + 2 IF/ELSE + WHILE

- ✓ **WHILE** alarm rings → snooze. 2 decisions (time check + bag packed). Python pseudocode: correct while/if/else syntax. All flowchart shapes used correctly.

PART 3 — AI · 10 MIN · ALL

AI always gives correct answers.

F

AI learns from biased data.

T

Safe to share passwords with AI.

F

Verify AI answers with trusted source.

T

AI can sound true but be made up.

T

Good Robot Habits: verify, ask adult, use as tool, think critically. **Bad:** trust blindly, share private info, copy without checking.

- ✓ **AI Scenario 1:** pet-name password = NOT safe (guessable & private info). **AI Scenario 2:** AI essay facts = must verify with book or trusted site first — AI can hallucinate.

EXIT TICKET · 5 MIN · ALL

- ✓ **Algorithm:** "Precise, ordered steps a computer follows exactly — like a recipe with no gaps."

- ✓ **Safety rule:** any 1 of the 5 Golden Rules in own words.

- ✓ **Can AI be wrong?** Yes — it can hallucinate (invent false facts that sound true). Always check a trusted source.

- ✓ **Biggest takeaway:** open-ended drawing/sentence — award for any genuine connection to lesson.

AI learns from data with possible bias.

T

Check AI answers with trusted source.

T

Share home address with AI chatbot.

F

AI can sound true but be made up.

T

- ✓ **3 Personal AI Safety Rules sample:** (1) Always verify AI facts. (2) Never share private info with AI. (3) If AI seems wrong, ask a trusted adult.

Responsible AI badge: title + ≥1 AI principle shown visually = full credit.

SECTION 4 — VOCAB · ~5 MIN · B & C

Algorithm

- ✓ "Precise ordered steps like a recipe." Sentence: "I follow an algorithm to brush my teeth."

Debug

- ✓ "Find and fix errors in code." Sentence: "I debugged my algorithm when my brother got confused."

Private Info

- ✓ "Personal data only trusted people should know." e.g. "My address is private."

AI

- ✓ "Programs that learn from data to make predictions." Analogy: "Fast pattern-spotter that can still be wrong."

REFLECTION + PARENT SIGN-OFF

- ✓ **Most interesting & question:** open-ended — all genuine reflections valid.

- ✓ **Drawing:** any sketch connecting today's lesson = full credit.

- ✓ **Parent sign-off:** signature + date required for completion credit.

Parents: Ask your child to teach you their algorithm — great dinner conversation!